

CSA0904-JAVA PROGRAMMING

COLLECTION-1

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1. Employee Name List (ArrayList)

Scenario

A company HR department wants to store the names of employees who attended a training program.

Question

Write a Java program to:

Create an ArrayList<String>

Add employee names: Ramesh, Suresh, Priya, Anu, Karthik

Display all employee names in the order they were added.

Code:

```
import java.util.ArrayList;

public class EmployeeList {

    public static void main(String[] args) {

        ArrayList<String> emp = new ArrayList<>();

        emp.add("Ramesh");
        emp.add("Suresh");
        emp.add("Priya");
        emp.add("Anu");
        emp.add("Karthik");

        System.out.println("Employee Names:");
        for (String name : emp) {
```

```

        System.out.println(name);
    }
}
}

```

Output:

The screenshot shows a Java IDE with a file named `EmployeeList.java`. The code defines a `public class EmployeeList` with a `main` method. Inside `main`, an `ArrayList<String>` named `emp` is created and populated with the names "Ramesh", "Suresh", "Priya", "Anu", and "Karthik". A `for` loop iterates over the `emp` list, printing each name. The output window on the right shows the execution results: "Employee Names:" followed by each name on a new line, and a final message "=== Code Execution Successful ===".

```

1 import java.util.ArrayList;
2
3 public class EmployeeList {
4     public static void main(String[] args) {
5         ArrayList<String> emp = new ArrayList<>();
6
7         emp.add("Ramesh");
8         emp.add("Suresh");
9         emp.add("Priya");
10        emp.add("Anu");
11        emp.add("Karthik");
12
13        System.out.println("Employee Names:");
14        for (String name : emp) {
15            System.out.println(name);
16        }
17    }
18 }

```

Output:

```

Employee Names:
Ramesh
Suresh
Priya
Anu
Karthik

=== Code Execution Successful ===

```

2.Restaurant Order Management (ArrayList)

Scenario

A restaurant stores the food items ordered by a customer.

Question

Write a Java program to:

Create an `ArrayList<String>`

Add items: Dosa, Idli, Pongal, Coffee

Remove Pongal

Display the updated order list and the total number of items.

Code:

```
import java.util.ArrayList;

public class RestaurantOrder {

    public static void main(String[] args) {

        ArrayList<String> order = new ArrayList<>();

        order.add("Dosa");
        order.add("Idli");
        order.add("Pongal");
        order.add("Coffee");
        order.remove("Pongal");

        System.out.println("Order List: " + order);
        System.out.println("Total Items: " + order.size());
    }
}
```

Output:



The screenshot shows a Java IDE with a file named 'RestaurantOrder.java'. The code is as follows:

```
1- import java.util.ArrayList;
2
3- public class RestaurantOrder {
4-     public static void main(String[] args) {
5-         ArrayList<String> order = new ArrayList<>();
6
7-         order.add("Dosa");
8-         order.add("Idli");
9-         order.add("Pongal");
10        order.add("Coffee");
11
12        order.remove("Pongal");
13
14        System.out.println("Order List: " + order);
15        System.out.println("Total Items: " + order.size());
16    }
17 }
```

The output window on the right shows the following results:

```
Order List: [Dosa, Idli, Coffee]
Total Items: 3

=== Code Execution Successful ===
```

3. Online Course Enrollment (ArrayList)

Scenario

An online learning platform stores the names of students enrolled in a Java course.

Question

Write a Java program to:

Store 5 student names in an ArrayList

Check whether Meena is enrolled

Print Enrolled if the name exists; otherwise print Not Enrolled.

Code:

```
import java.util.ArrayList;

public class CourseEnrollment {

    public static void main(String[] args) {

        ArrayList<String> students = new ArrayList<>();

        students.add("Ravi");
        students.add("Meena");
        students.add("Anu");
        students.add("Karthik");
        students.add("Priya");

        if (students.contains("Meena")) {

            System.out.println("Enrolled");

        } else {

            System.out.println("Not Enrolled");

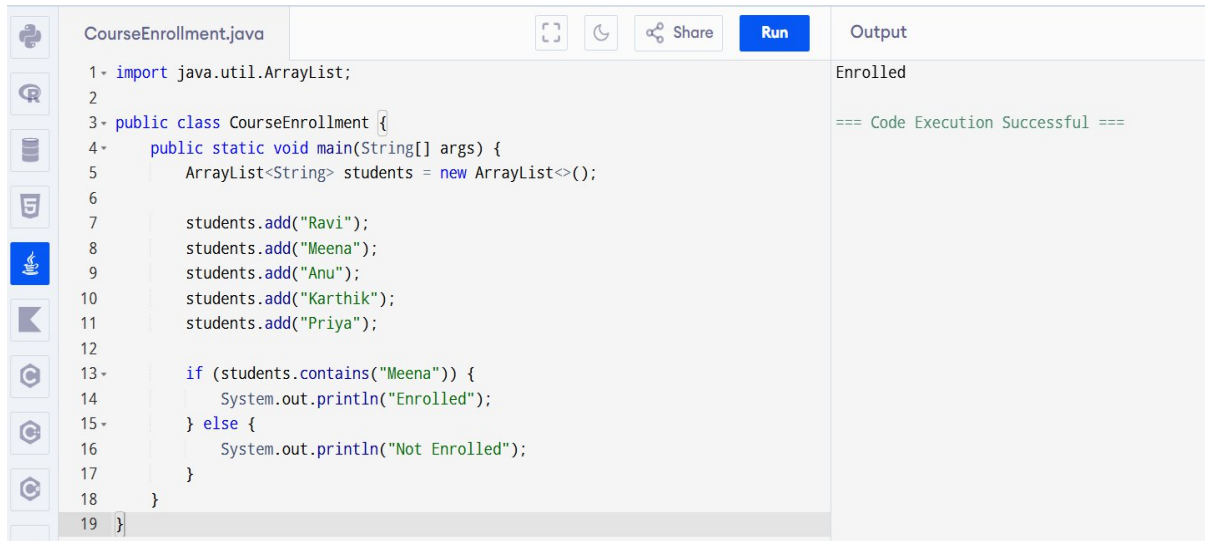
        }

    }

}
```

}

Output:



```
CourseEnrollment.java
1 import java.util.ArrayList;
2
3 public class CourseEnrollment {
4     public static void main(String[] args) {
5         ArrayList<String> students = new ArrayList<>();
6
7         students.add("Ravi");
8         students.add("Meena");
9         students.add("Anu");
10        students.add("Karthik");
11        students.add("Priya");
12
13        if (students.contains("Meena")) {
14            System.out.println("Enrolled");
15        } else {
16            System.out.println("Not Enrolled");
17        }
18    }
19 }
```

Output

Enrolled

=== Code Execution Successful ===

4.Unique Email Registration (Set)

Scenario

A website should not allow duplicate email registrations.

Question

Write a Java program to:

Create a `HashSet<String>`

Add emails: a@gmail.com, b@gmail.com, a@gmail.com, c@gmail.com

Display all unique email addresses

Display the total number of unique registered users.

Code:

```
import java.util.HashSet;
```

```
public class EmailRegistration {
```

```
    public static void main(String[] args) {
```

```
        HashSet<String> emails = new HashSet<>();
```

```

        emails.add("a@gmail.com");
        emails.add("b@gmail.com");
        emails.add("a@gmail.com");
        emails.add("c@gmail.com");

        System.out.println("Unique Emails: " + emails);
        System.out.println("Total Users: " + emails.size());
    }
}

```

Output:

File	Code	Output
EmailRegistration.java	<pre> 1 import java.util.HashSet; 2 3 public class EmailRegistration { 4 public static void main(String[] args) { 5 HashSet<String> emails = new HashSet<>(); 6 7 emails.add("a@gmail.com"); 8 emails.add("b@gmail.com"); 9 emails.add("a@gmail.com"); 10 emails.add("c@gmail.com"); 11 12 System.out.println("Unique Emails: " + emails); 13 System.out.println("Total Users: " + emails.size()); 14 } 15 } </pre>	<pre> Unique Emails: [b@gmail.com, c@gmail.com, a@gmail.com] Total Users: 3 === Code Execution Successful === </pre>

5.Library Book ID Tracking (Set)

Scenario

A library stores book ID numbers. The same book ID should not be entered twice.

Question

Write a Java program to:

Create a `HashSet<Integer>`

Add book IDs: 101, 102, 103, 101, 104

Display all unique book IDs

Print Duplicate Book ID Found when a duplicate ID is added.

Code:

```
import java.util.HashSet;

public class LibraryBookID {

    public static void main(String[] args) {

        HashSet<Integer> books = new HashSet<>();

        int[] ids = {101, 102, 103, 101, 104};

        for (int id : ids) {
            if (!books.add(id)) {
                System.out.println("Duplicate Book ID Found");
            }
        }

        System.out.println("Unique Book IDs: " + books);
    }
}
```

Output:

	LibraryBookID.java <pre>1 import java.util.HashSet; 2 3 public class LibraryBookID { 4 public static void main(String[] args) { 5 HashSet<Integer> books = new HashSet<>(); 6 7 int[] ids = {101, 102, 103, 101, 104}; 8 9 for (int id : ids) { 10 if (!books.add(id)) { 11 System.out.println("Duplicate Book ID Found"); 12 } 13 } 14 15 System.out.println("Unique Book IDs: " + books); 16 } 17 }</pre>	Output <pre>Duplicate Book ID Found Unique Book IDs: [101, 102, 103, 104] === Code Execution Successful ===</pre>
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